

UNIVERSITY OF RUHUNA
FACULTY OF TECHNOLOGY

ICT2232 – SOFTWARE ENGINEERING

MINI PROJECT PROPOSAL

PUBLIC SERVICE COMPLAINT & RESOLUTION SYSTEM

Group No: 15

Group Members:

TG/2023/1761

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1.Project Overview

The Public Service Complaint & Resolution System is a proposed web-based software solution designed to provide a centralised platform for citizens to report public service problems, monitor complaint progress and receive resolution updates. The system will also support authorised public service officers in reviewing, assigning, prioritising, updating and resolving complaints.

The project is developed for ICT2232 Software Engineering and will demonstrate the early stages of the Software Development Life Cycle (SDLC). This proposal covers Stage 1 (Problem & Project Definition), Stage 2 (Process Planning), Stage 3 (Agile Planning) and Stage 4 (Requirements Engineering), corresponding to Milestones 1–3.

2. Stage 1 – Problem & Project Definition

2.1 Background

Public services such as waste management, road maintenance, water supply, sanitation, public transportation and other community services directly affect citizens. When a service problem occurs, citizens need a convenient way to report the issue and obtain information about how it is being handled.

Traditional complaint channels may include telephone calls, physical visits, emails and informal communication. These channels can make it difficult for citizens to track complaints and for service organizations to maintain consistent records, assign responsibility, prioritize urgent cases and monitor unresolved complaints.

The proposed system addresses this gap by providing a structured digital complaint lifecycle: submission, review, assignment, prioritization, progress updates, resolution and citizen feedback.

2.2 Problem Statement

Citizens may experience difficulty reporting public service issues and following up on complaints because complaint information is often handled through separate or manual communication channels. This can result in limited visibility of complaint status, delays in communication, difficulty identifying responsible officers and challenges in monitoring unresolved complaints.

Public service organizations also require an organized mechanism to record, categorize, prioritize, assign and monitor complaints. Therefore, a centralized Public Service Complaint & Resolution System is required to improve accessibility, transparency, communication and efficiency in complaint management.

2.3 Proposed Solution

The proposed system will provide a role-based web application through which citizens can submit and track complaints, while authorized public service officers can manage the complaint lifecycle. Administrators will manage users, service categories and system-level information.

- Citizens can register, log in, submit complaints and view complaint history.
- Citizens can provide complaint details such as category, description and location.
- Citizens can upload supporting evidence where applicable.
- Officers can review, assign, prioritize and update complaints.
- Officers can record progress remarks and resolution information.
- Citizens can receive complaint progress and resolution updates.
- Citizens can provide feedback after a complaint is resolved.
- Administrators can manage users, complaint categories and monitoring information.

2.4 Stakeholders and Needs

- Citizens
  Easy complaint submission, tracking, status visibility, updates and feedback.
- Public Service Officers
  Centralized complaints, search/filtering, assignment, prioritization and status management.
- Department Managers
  Visibility of department complaints, workload and unresolved cases.
- System Administrator
  User management, category management, monitoring and secure system administration.
- Public Service Organizations
  Improved accountability, organized records and better monitoring of service issues.
- Community
  More transparent and responsive handling of public service problems.

2.5 Project Objectives

1. Provide citizens with a convenient digital platform to submit public service complaints.
2. Allow citizens to track the status and progress of submitted complaints.
3. Enable authorized officers to manage, assign, prioritize and resolve complaints.
4. Maintain accurate and centralized complaint records.
5. Provide timely progress and resolution updates to citizens.
6. Provide a feedback mechanism after complaint resolution.
7. Support administrators in managing users and complaint categories.
8. Improve transparency, accountability and efficiency in complaint handling.
9. Apply appropriate authentication and role-based authorization.
10. Develop a system that can be extended with additional public services in the future.

2.6 Project Scope

2.6.1 In-Scope

- Citizen registration, login and profile management.
- Role-based authentication for citizens, officers and administrators.
- Complaint submission and validation.
- Complaint category and priority management.
- Complaint description and location information.
- Supporting image/document upload.
- Complaint assignment to relevant officers/departments.
- Complaint status and progress updates.
- Resolution details and complaint history.
- Notifications or in-system updates.
- Citizen feedback after resolution.
- Complaint search and filtering.
- Administrative user and category management.
- Basic monitoring/dashboard information.

2.6.2 Out-of-Scope

- Online payment processing.
- Real-time GPS tracking of field officers.
- Automatic legal case management.
- AI-based automatic complaint classification.
- Direct integration with every government information system.
- Emergency response and disaster management.
- Automated physical inspection or field-service dispatch.

2.7 Assumptions

- Users have access to an internet-connected device and web browser.
- Citizens provide truthful and sufficient complaint information.
- Authorized officers have access to the system and update complaint information.
- Public service categories and departments are maintained as system data.
- Uploaded evidence follows supported file types and size limits.
- Prototype demonstrations will use fictional or sample data.
- External government system integration is not required for the initial prototype.

2.8 Constraints

- Limited academic project duration.

- Limited development resources and team size.
- Limited access to real government databases and services.
- Possible limitations in implementing real-time external notifications.
- Security and privacy requirements must be considered despite the prototype nature.
- The system cannot represent every complaint workflow used by all public organizations.

3. Stage 2 – Process Planning

3.1 Selected Process Model

The project will follow an Agile software development approach using Scrum practices.

3.2 Process Model Justification

- Requirements can be refined as the team receives feedback from stakeholders and lecturer reviews.
- The system can be developed incrementally, allowing working features to be demonstrated early.
- Short sprints make it easier to identify defects and make corrections before later development.
- Priorities can be adjusted through the product backlog.
- Continuous testing can be integrated with development.
- The approach is suitable for a small student development team.

3.3 Development Approach

Each sprint will produce a usable increment. The general cycle will be:

Sprint Planning → Design → Implementation → Testing → Sprint Review → Feedback → Backlog Refinement → Next Sprint

3.4 Initial Project Plan

Phase	Main Activities	Expected Output
Project Definition	Problem, stakeholders, objectives, scope, assumptions and constraints.	Approved project definition.
Process Planning	Select and justify Agile/Scrum; prepare initial development plan.	Process model and project plan.
Agile Planning	Create backlog, user stories, priorities, acceptance criteria and sprint plan.	Agile planning baseline.
Requirements Engineering	Define FRs, NFRs, use cases and traceability.	Requirements baseline.
Design	Architecture, UML, database and UI design.	Design baseline.

Implementation	Develop prioritized modules and integrate components.	Working software.
Testing	Unit, integration, system and acceptance testing.	Test evidence and corrected defects.
Deployment	Configure and deploy the application.	Demonstrable deployed system.
Maintenance	Analyze and implement lecturer-approved change request.	Updated artefacts and system.
Security	Threat assessment and implementation of controls.	Security evidence.

4. Stage 3 – Agile Planning

4.1 Product Backlog

Must Have

- User registration
- Secure login/logout
- Role-based access control
- Submit complaint
- Complaint category selection
- View submitted complaints
- Track complaint status
- Assign complaint
- Set complaint priority
- Update complaint progress
- Resolve complaint
- Complaint notifications/updates
- Admin user management

Should Have

- Citizen profile management
- Upload supporting evidence
- Search and filter complaints
- Citizen feedback
- Complaint history
- Complaint category management

Could Have

- Dashboard/reports

4.2 User Stories

Must Have

- User registration
- Secure login
- Submit complaint
- Complaint category selection
- Track complaint
- Receive complaint updates
- View assigned complaints
- Update complaint status
- Prioritize complaints
- Add progress remarks
- Mark complaint as resolved
- Manage users

Should Have

- Upload supporting evidence
- Provide feedback after resolution
- Manage complaint categories

4.3 Acceptance Criteria

Submit Complaint

- Citizen is authenticated.
- Required complaint fields are validated.
- A unique complaint ID is generated.
- Complaint is saved in the database.
- Initial status is set to Submitted.
- Confirmation is shown to the citizen.

Track Complaint

- Citizen can access their submitted complaints.
- Complaint ID and category are displayed.
- Current status is displayed.
- Progress information is displayed when available.
- Resolution information is displayed for resolved complaints.

Update Status

- Only an authorized officer can update the status.

- A valid status transition is selected.
- New status is stored.
- Complaint history is updated.
- Citizen can view the updated status.

Resolve Complaint

- Officer can provide resolution information.
- Status changes to Resolved.
- Resolution date is stored.
- Citizen receives/sees a resolution update.
- Feedback becomes available where implemented.

Manage Users

- Only an administrator can access user management.
- Administrator can view permitted user records.
- Administrator can activate/deactivate or update permitted user information.
- Unauthorized users are denied access.

4.4 Sprint Plan

1. Sprint 1 - Foundation & Authentication

   Project setup, database foundation, registration, login, roles and basic UI.

2. Sprint 2 - Complaint Submission

   Complaint form, categories, validation, evidence and complaint record creation.

3. Sprint 3 - Complaint Processing

   Officer dashboard, assignment, priority, status and progress remarks.

4. Sprint 4 - Tracking & Resolution

   Citizen tracking, notifications/updates, resolution and feedback.

5. Sprint 5 - Quality & Release

   Integration, system testing, security, defect fixing and deployment preparation.

4.5 Task Allocation

Member	Main Area	Main Responsibilities
TG/2023/1731	Requirements & Citizen Module	Stakeholder analysis, requirements, user stories, citizen registration, login, profile and complaint submission
TG/2023/1710	Complaint Management Module	Complaint categories, complaint details, evidence upload, complaint assignment and priority management
TG/2023/1757	Officer Module	Officer dashboard, complaint review, status updates, progress remarks and resolution management
TG/2023/1761	Admin & Database	Database/ER design, user management, category management, admin dashboard and database integration
TG/2023/1776	Quality, Security & Deployment	Test planning, system/integration testing, security controls, error handling, deployment and maintenance change request

5. Stage 4 – Requirements Engineering

5.1 Functional Requirements

- The system shall allow citizens to create an account.
- The system shall allow registered users to log in and log out.
- The system shall enforce role-based access to protected functions.
- The system shall allow citizens to submit public service complaints
- The system shall allow citizens to select a complaint category.
- The system shall allow citizens to provide a complaint description and relevant location
- The system shall allow citizens to attach supported evidence where applicable
- The system shall generate a unique identifier for each complaint.
- The system shall allow authorized officers to view and manage complaints.
- The system shall allow authorized officers to assign complaints to responsible officers/departments.
- The system shall allow authorized officers to set complaint priority.
- The system shall allow authorized officers to update complaint status.
- The system shall record progress remarks and resolution information.
- The system shall allow citizens to track their complaints.
- The system shall provide complaint progress and resolution updates.
- The system shall maintain complaint history.
- The system shall allow citizens to provide feedback after resolution.
- The system shall allow administrators to manage users.
- The system shall allow administrators to manage complaint categories.
- The system shall provide administrators with complaint monitoring information.

5.2 Non-Functional Requirements

- Performance
  Common operations should respond within an acceptable time under normal project usage
- Usability
  The interface should be simple and understandable for citizens and officers.
- Security
  Protected functions shall require authenticated access.
- Authorization
  Users shall only access functions permitted by their roles.
- Reliability
  Complaint records should be stored accurately and consistently.
- Availability
  The deployed system should be accessible during the intended demonstration/usage period.
- Maintainability
  The software should use modular and understandable code to support future changes
- Scalability
  The system should allow additional categories, departments and users to be added.
- Data Integrity
  Input validation shall prevent invalid or incomplete records.
- Privacy
  Personal and complaint information shall only be accessible to authorized users.

5.3 Actors

- Citizen – Submits complaints, tracks progress, receives updates and provides feedback.
- Public Service Officer – Reviews, assigns, prioritizes, updates and resolves complaints.
- Administrator – Manages users, categories and system-level monitoring information.

5.4 Important Use Cases

- Register Account
  Citizen creates a new account.
- Login
  User authenticates and accesses permitted functions.
- Submit Complaint
  Citizen enters complaint information and submits it.
- Track Complaint
  Citizen views the current status and progress of a complaint.
- Manage Complaint
  Officer reviews, assigns, and prioritizes a complaint.

- Update Complaint
  Officer records progress and changes complaint status
- Resolve Complaint
  Officer records resolution and closes the processing stage.
- Provide Feedback
  Citizen provides feedback after resolution.
- Manage Users
  Administrator manages user access and permitted user information.
- Manage Categories
  Administrator creates and updates complaint categories.

5.5 Use Case Descriptions

Submit Complaint

- Primary Actor: Citizen
- Precondition: Citizen is logged in.
- Main Flow:
  1. Open complaint form.
  2. Enter required details.
  3. Select category.
  4. Add location/evidence if applicable.
  5. Submit complaint.
- Post condition: System validates the data, creates a complaint ID, and stores the complaint with Submitted status.
- Alternative/Exception: Required information is missing or invalid.

Track Complaint

- Primary Actor: Citizen
- Precondition: Citizen has at least one complaint.
- Main Flow:
  1. Open My Complaints.
  2. Select a complaint.
  3. View status and progress.
- Post condition: System displays complaint details, current status, and available updates.
- Alternative/Exception: Complaint does not belong to the citizen or does not exist.

Manage Complaint

- Primary Actor: Officer
- Precondition: Officer is authenticated and authorized.
- Main Flow:
  1. View complaint queue.
  2. Select a complaint.

3. Review details.
4. Assign responsible officer/department.
5. Set priority.

- Post condition: Assignment and priority are saved.
- Alternative/Exception: Officer is not authorized or required data is invalid.

Update Complaint

- Primary Actor: Officer
- Precondition: Officer has access to the complaint.
- Main Flow:
    1. Open complaint.
    2. Add progress remark.
    3. Select valid status.
    4. Save.
- Post condition: New status and progress information are stored and visible to permitted users.
- Alternative/Exception: Invalid status or unauthorized update.

Resolve Complaint

- Primary Actor: Officer
- Precondition: Complaint work has been completed.
- Main Flow:
    1. Open complaint.
    2. Enter resolution details.
    3. Mark as resolved.
    4. Save.
- Post condition: Resolution details and date are recorded, and the citizen can see the resolution update.
- Alternative/Exception: Resolution information is incomplete

5.6 Prioritization

Must Have

- The system shall allow citizens to create an account.
- The system shall allow registered users to log in and log out.
- The system shall enforce role-based access to protected functions.
- The system shall allow citizens to submit public service complaints.
- The system shall allow citizens to select a complaint category.

- The system shall allow citizens to provide a complaint description and relevant location.
- The system shall generate a unique identifier for each complaint.
- The system shall allow authorized officers to view and manage complaints.
- The system shall allow authorized officers to assign complaints to responsible officers/departments.
- The system shall allow authorized officers to set complaint priority.
- The system shall allow authorized officers to update complaint status.
- The system shall record progress remarks and resolution information.
- The system shall allow citizens to track their complaints.
- The system shall provide complaint progress and resolution updates.
- The system shall allow administrators to manage users.

Should Have

- The system shall allow citizens to attach supported evidence where applicable.
- The system shall maintain complaint history.
- The system shall allow citizens to provide feedback after resolution.
- The system shall allow administrators to manage complaint categories.

Could Have

- The system shall provide administrators with complaint monitoring information.
- Enhanced dashboard and reporting features could be included to provide additional monitoring and analytical information.

Won't Have – Initial Release

- AI-based complaint classification.
- GPS-based officer tracking.
- Online payments.
- Emergency response services.
- Broad external government system integrations.

5.7 Requirements Traceability Matrix

Requirement	Planned Design	Implementation Area

Registration	User/Authentication Design	Registration Module
Login/logout	Authentication Design	Authentication Module
Role access	Security Design	Authorisation Module
Submit complaint	Complaint Module	Complaint Service/Form
Category	Complaint Module	Category Selection
Unique complaint ID	Database/Complaint Design	Complaint Service
Assignment	Officer Module	Assignment Function
Priority	Officer Module	Priority Function
Status update	Complaint Workflow	Status Function
Tracking	Citizen Dashboard	Tracking Function
Updates	Notification/Update Design	Notification Function
Feedback	Feedback Module	Feedback Function
User management	Admin Module	User Management
Security	Security Architecture	Authentication/Controls
Authorization	Security Architecture	Role-based Access
Input validation	Validation Design	Validation Layer

6. Milestone Mapping

Milestone 1 – Project Definition & Process Model

Background, problem statement, stakeholders, objectives, scope, assumptions, constraints, Agile/Scrum selection and project plan.

Milestone 2 – Agile Planning

Product backlog, user stories, priorities, acceptance criteria, sprint plan and development approach.

Milestone 3 – Requirements Baseline

Functional requirements, non-functional requirements, actors, use cases, prioritization and requirements traceability.

7. Conclusion

The Public Service Complaint & Resolution System provides a practical software engineering solution for improving the submission, management, tracking and resolution of public service complaints. The proposal establishes the problem domain, stakeholder needs, objectives, scope and constraints, and selects an Agile Scrum-based process to support iterative development.

The product backlog, user stories, acceptance criteria and sprint plan provide the foundation for Agile development, while the functional and non-functional requirements, use cases, prioritisation and traceability matrix establish the initial requirements baseline. These outputs will directly inform the subsequent design, implementation, testing, deployment, maintenance and security stages of the project.